

Foundation (Jalapeno)

- Q1.** Which of the following is a condition for a function to have an inverse?
- (A) The function is one-to-one
 - (B) The function is many-to-one
 - (C) The function is onto
 - (D) The function is even
 - (E) The function is odd
- Q2.** If $f(x) = x^2$, what is the inverse function?
- (A) $f^{-1}(x) = \sqrt{x}$
 - (B) $f^{-1}(x) = -\sqrt{x}$
 - (C) $f^{-1}(x) = \frac{1}{x}$
 - (D) $f^{-1}(x) = \frac{1}{\sqrt{x}}$
 - (E) The function does not have an inverse
- Q3.** What is the purpose of verifying inverses using composition?
- (A) To check if the function is one-to-one
 - (B) To check if the function is onto
 - (C) To check if the inverse is a function
 - (D) To check if $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$
 - (E) To graph the function
- Q4.** If $g(x) = 2x - 3$, what is the inverse function?
- (A) $g^{-1}(x) = \frac{x+3}{2}$
 - (B) $g^{-1}(x) = \frac{x-3}{2}$
 - (C) $g^{-1}(x) = \frac{x+3}{2}$
 - (D) $g^{-1}(x) = \frac{x}{2} + 3$
 - (E) $g^{-1}(x) = 2x + 3$
- Q5.** Which of the following statements is true about inverse functions?
- (A) The domain of the inverse is equal to the range of the original function
 - (B) The range of the inverse is equal to the domain of the original function
 - (C) Both A and B
 - (D) Neither A nor B
 - (E) The inverse function is always one-to-one
- Q6.** If $h(x) = x^3 + 1$, what is the inverse function?
- (A) $h^{-1}(x) = \sqrt[3]{x-1}$
 - (B) $h^{-1}(x) = \sqrt[3]{x+1}$
 - (C) $h^{-1}(x) = \frac{1}{x^3}$
 - (D) $h^{-1}(x) = x^3 - 1$
 - (E) $h^{-1}(x) = \frac{1}{x}$
- Q7.** What is an example of a real-world context where inverse functions are used?
- (A) Calculating the area of a circle
 - (B) Finding the equation of a line
 - (C) Determining the pH of a solution
 - (D) Modeling population growth
 - (E) Encrypting data
- Q8.** If $f(x) = \frac{1}{x}$, what is the inverse function?
- (A) $f^{-1}(x) = \frac{1}{x}$
 - (B) $f^{-1}(x) = x$
 - (C) $f^{-1}(x) = -x$
 - (D) $f^{-1}(x) = \frac{1}{x^2}$
 - (E) $f^{-1}(x) = x^2$

Open Ended Questions (FRQ)

- Q9.** Find the inverse of the function $f(x) = 3x - 2$. Verify your answer using composition.
- Q10.** A company produces widgets, and the cost C (in dollars) is related to the number of widgets x produced. The cost function is given by $C(x) = 2x + 5$. Find the inverse function and interpret its meaning in the context of the problem.

Chef's Tips

- Make sure to check the work for questions 9 and 10
- Allow students to use calculators for verification
- Provide feedback on the correctness of the inverse functions